Practical -1

DATABASE SYSTEMS LAB ASSIGNMENT REPORT

Course Title: Advanced Database Management Systems

Topic: Relational Schema Design, Integrity Constraints, and Advanced SQL Queries

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1. Executive Summary & Design Overview

This report demonstrates a production-grade relational schema implementing comprehensive integrity constraints. The target architecture maps an enterprise order-fulfillment system spanning 14 tables.

Core Architecture Metrics

- **Total Component Tables:** 14 structured entities.
- **Minimum Table Width:** Every table contains 8 or more verified attributes.
- **Foreign Key Connectors:** 15 explicit active relationships map data flow across dependencies.

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2. Relational Schema Blueprint (DDL Script)

```
-- ==========================================
-- 1. DROP TABLES (For clean re-runs)
-- ==========================================
DROP TABLE IF EXISTS Inventory;
DROP TABLE IF EXISTS Warehouses;
DROP TABLE IF EXISTS ReviewComments;
DROP TABLE IF EXISTS OrderDetails;
DROP TABLE IF EXISTS Shipments;
DROP TABLE IF EXISTS Payments;
DROP TABLE IF EXISTS Orders;
DROP TABLE IF EXISTS ProductSuppliers;
DROP TABLE IF EXISTS Products;
DROP TABLE IF EXISTS Categories;
DROP TABLE IF EXISTS Employees;
DROP TABLE IF EXISTS Departments;
DROP TABLE IF EXISTS Suppliers;
DROP TABLE IF EXISTS Customers;

-- ==========================================
```

```sql
-- 2. CREATE TABLES WITH CONSTRAINTS (8+ Attributes each)
-- ============================================

CREATE TABLE Customers (
    CustomerID INT PRIMARY KEY,
    CustomerName VARCHAR(100) NOT NULL,
    Email VARCHAR(100) UNIQUE NOT NULL,
    Phone VARCHAR(20) NOT NULL,
    Address VARCHAR(255) NOT NULL,
    City VARCHAR(100) NOT NULL,
    Country VARCHAR(100) DEFAULT 'India',
    PostalCode VARCHAR(20) NOT NULL,
    IsActive INT DEFAULT 1 CHECK (IsActive IN (0, 1))
);

CREATE TABLE Suppliers (
    SupplierID INT PRIMARY KEY,
    SupplierName VARCHAR(100) NOT NULL,
    ContactName VARCHAR(100) NOT NULL,
    Email VARCHAR(100) UNIQUE NOT NULL,
    Phone VARCHAR(20) NOT NULL,
    Address VARCHAR(255) NOT NULL,
    City VARCHAR(100) NOT NULL,
    Country VARCHAR(100) DEFAULT 'India',
    SupplierStatus VARCHAR(20) DEFAULT 'Active' CHECK (SupplierStatus IN ('Active',
'Inactive'))
);

CREATE TABLE Departments (
    DepartmentID INT PRIMARY KEY,
    DepartmentName VARCHAR(100) UNIQUE NOT NULL,
    ManagerName VARCHAR(100) NOT NULL,
    Budget DECIMAL(12,2) CHECK (Budget >= 0),
    Location VARCHAR(100) NOT NULL,
    PhoneExtension VARCHAR(10) NOT NULL,
    CreatedDate DATE DEFAULT CURRENT_DATE,
    IsOperational INT DEFAULT 1 CHECK (IsOperational IN (0, 1))
);

CREATE TABLE Employees (
    EmployeeID INT PRIMARY KEY,
    FirstName VARCHAR(50) NOT NULL,
    LastName VARCHAR(50) NOT NULL,
    Email VARCHAR(100) UNIQUE NOT NULL,
    Phone VARCHAR(20) NOT NULL,
    HireDate DATE DEFAULT CURRENT_DATE,
    Salary DECIMAL(10,2) DEFAULT 25000 CHECK (Salary >= 10000),
```

```sql
    EmployeeStatus VARCHAR(20) DEFAULT 'Active' CHECK (EmployeeStatus IN ('Active',
'On Leave', 'Terminated')),
    DepartmentID INT,
    CONSTRAINT fk_emp_dept FOREIGN KEY (DepartmentID) REFERENCES
Departments(DepartmentID) ON DELETE SET NULL ON UPDATE CASCADE
);

CREATE TABLE Categories (
    CategoryID INT PRIMARY KEY,
    CategoryName VARCHAR(100) UNIQUE NOT NULL,
    Description TEXT,
    TaxRate DECIMAL(4,2) DEFAULT 18.00 CHECK (TaxRate >= 0),
    CreatedDate DATE DEFAULT CURRENT_DATE,
    MinStockThreshold INT DEFAULT 5 CHECK (MinStockThreshold >= 0),
    MaxStockThreshold INT DEFAULT 500 CHECK (MaxStockThreshold >
MinStockThreshold),
    IsDiscontinued INT DEFAULT 0 CHECK (IsDiscontinued IN (0, 1))
);

CREATE TABLE Products (
    ProductID INT PRIMARY KEY,
    ProductName VARCHAR(100) NOT NULL,
    SKU VARCHAR(50) UNIQUE NOT NULL,
    Price DECIMAL(10,2) NOT NULL CHECK (Price > 0),
    Discount DECIMAL(5,2) DEFAULT 0 CHECK (Discount >= 0 AND Discount <= 100),
    Stock INT DEFAULT 0 CHECK (Stock >= 0),
    ReviewRating INT DEFAULT 3 CHECK (ReviewRating BETWEEN 1 AND 5),
    CreatedDate DATE DEFAULT CURRENT_DATE,
    CategoryID INT NOT NULL,
    CONSTRAINT fk_prod_cat FOREIGN KEY (CategoryID) REFERENCES
Categories(CategoryID) ON DELETE RESTRICT ON UPDATE CASCADE
);

CREATE TABLE ProductSuppliers (
    ProductID INT,
    SupplierID INT,
    SupplyPrice DECIMAL(10,2) NOT NULL CHECK (SupplyPrice > 0),
    LeadTimeDays INT DEFAULT 7 CHECK (LeadTimeDays >= 0),
    IsPrimarySupplier INT DEFAULT 1 CHECK (IsPrimarySupplier IN (0, 1)),
    ContractStartDate DATE NOT NULL,
    ContractEndDate DATE,
    LastOrderDate DATE,
    PRIMARY KEY (ProductID, SupplierID),
    CONSTRAINT fk_ps_prod FOREIGN KEY (ProductID) REFERENCES
Products(ProductID) ON DELETE CASCADE,
    CONSTRAINT fk_ps_supp FOREIGN KEY (SupplierID) REFERENCES
Suppliers(SupplierID) ON DELETE CASCADE
);
```

```sql
CREATE TABLE Orders (
   OrderID INT PRIMARY KEY,
   OrderDate DATE DEFAULT CURRENT_DATE,
   RequiredDate DATE NOT NULL,
   OrderStatus VARCHAR(20) DEFAULT 'Pending' CHECK (OrderStatus IN ('Pending',
'Processing', 'Shipped', 'Delivered', 'Cancelled')),
   TotalAmount DECIMAL(12,2) DEFAULT 0.00 CHECK (TotalAmount >= 0),
   CustomerID INT,
   EmployeeID INT,
   ShippingMethod VARCHAR(50) DEFAULT 'Standard',
   CONSTRAINT fk_ord_cust FOREIGN KEY (CustomerID) REFERENCES
Customers(CustomerID) ON DELETE SET NULL,
   CONSTRAINT fk_ord_emp FOREIGN KEY (EmployeeID) REFERENCES
Employees(EmployeeID) ON DELETE NO ACTION
);

CREATE TABLE Payments (
   PaymentID INT PRIMARY KEY,
   PaymentDate DATE DEFAULT CURRENT_DATE,
   PaymentMethod VARCHAR(50) NOT NULL CHECK (PaymentMethod IN ('Credit Card',
'Debit Card', 'Net Banking', 'UPI', 'Cash')),
   PaymentStatus VARCHAR(20) DEFAULT 'Unpaid' CHECK (PaymentStatus IN ('Unpaid',
'Paid', 'Refunded', 'Failed')),
   AmountPaid DECIMAL(12,2) NOT NULL CHECK (AmountPaid >= 0),
   TransactionRef VARCHAR(100) UNIQUE,
   Remarks TEXT,
   OrderID INT NOT NULL,
   CONSTRAINT fk_pay_ord FOREIGN KEY (OrderID) REFERENCES Orders(OrderID) ON
DELETE CASCADE
);

CREATE TABLE Shipments (
   ShipmentID INT PRIMARY KEY,
   ShipmentDate DATE,
   EstimatedDeliveryDate DATE NOT NULL,
   ActualDeliveryDate DATE,
   CarrierName VARCHAR(100) NOT NULL,
   TrackingNumber VARCHAR(100) UNIQUE NOT NULL,
   ShippingCost DECIMAL(10,2) DEFAULT 0.00 CHECK (ShippingCost >= 0),
   ShipmentStatus VARCHAR(20) DEFAULT 'Manifest' CHECK (ShipmentStatus IN
('Manifest', 'In Transit', 'Out for Delivery', 'Delivered', 'Returned')),
   OrderID INT NOT NULL,
   CONSTRAINT fk_ship_ord FOREIGN KEY (OrderID) REFERENCES Orders(OrderID)
ON DELETE CASCADE
);

CREATE TABLE OrderDetails (
```

```
    OrderDetailID INT PRIMARY KEY,
    OrderID INT NOT NULL,
    ProductID INT NOT NULL,
    Quantity INT NOT NULL CHECK (Quantity > 0),
    UnitPrice DECIMAL(10,2) NOT NULL CHECK (UnitPrice > 0),
    Discount DECIMAL(5,2) DEFAULT 0 CHECK (Discount >= 0 AND Discount <= 100),
    TaxAmount DECIMAL(10,2) DEFAULT 0.00 CHECK (TaxAmount >= 0),
    LineTotal DECIMAL(12,2) NOT NULL CHECK (LineTotal >= 0),
    CONSTRAINT fk_od_ord FOREIGN KEY (OrderID) REFERENCES Orders(OrderID) ON
DELETE CASCADE,
    CONSTRAINT fk_od_prod FOREIGN KEY (ProductID) REFERENCES
Products(ProductID) ON DELETE RESTRICT
);

CREATE TABLE ReviewComments (
    ReviewID INT PRIMARY KEY,
    ProductID INT NOT NULL,
    CustomerID INT,
    Rating INT DEFAULT 3 CHECK (Rating BETWEEN 1 AND 5),
    ReviewComment TEXT,
    ReviewDate DATE DEFAULT CURRENT_DATE,
    IsVerifiedPurchase INT DEFAULT 1 CHECK (IsVerifiedPurchase IN (0, 1)),
    HelpfulVotes INT DEFAULT 0 CHECK (HelpfulVotes >= 0),
    CONSTRAINT fk_rev_prod FOREIGN KEY (ProductID) REFERENCES
Products(ProductID) ON DELETE CASCADE,
    CONSTRAINT fk_rev_cust FOREIGN KEY (CustomerID) REFERENCES
Customers(CustomerID) ON DELETE SET NULL
);

CREATE TABLE Warehouses (
    WarehouseID INT PRIMARY KEY,
    WarehouseName VARCHAR(100) NOT NULL,
    Location VARCHAR(100) NOT NULL,
    Capacity INT NOT NULL CHECK (Capacity > 0),
    ManagerName VARCHAR(100) NOT NULL,
    ContactNumber VARCHAR(20) NOT NULL,
    OperatingExpense DECIMAL(10,2) DEFAULT 5000.00 CHECK (OperatingExpense >= 0),
    IsClimateControlled INT DEFAULT 0 CHECK (IsClimateControlled IN (0, 1))
);

CREATE TABLE Inventory (
    InventoryID INT PRIMARY KEY,
    WarehouseID INT NOT NULL,
    ProductID INT NOT NULL,
    QuantityInHand INT DEFAULT 0 CHECK (QuantityInHand >= 0),
    BinLocation VARCHAR(50) NOT NULL,
    LastStockTakeDate DATE DEFAULT CURRENT_DATE,
    ReorderLevel INT DEFAULT 10 CHECK (ReorderLevel >= 0),
```

```
    IsQuarantined INT DEFAULT 0 CHECK (IsQuarantined IN (0, 1)),
    CONSTRAINT fk_inv_wh FOREIGN KEY (WarehouseID) REFERENCES
Warehouses(WarehouseID) ON DELETE CASCADE,
    CONSTRAINT fk_inv_prod FOREIGN KEY (ProductID) REFERENCES
Products(ProductID) ON DELETE CASCADE
);
```

Available Tables

Categories

CategoryID	CategoryName	Description	TaxRate	Crea
empty				

Customers

CustomerID	CustomerName	Email	Phone	Address
empty				

Departments

DepartmentID	DepartmentName	ManagerName	Budget
empty			

Employees

EmployeeID	FirstName	LastName	Email	Phone	H
empty					

Inventory

InventoryID	WarehouseID	ProductID	QuantityInHand

Inventory

InventoryID	WarehouseID	ProductID	QuantityInHand
empty			

OrderDetails

OrderDetailID	OrderID	ProductID	Quantity	UnitPrice
empty				

Orders

OrderID	OrderDate	RequiredDate	OrderStatus	TotalA
empty				

Payments

PaymentID	PaymentDate	PaymentMethod	PaymentStatus
empty			

Shipments

ShipmentID	ShipmentDate	EstimatedDeliveryDate	ActualD
empty			

Shippings

shipping_id	status	customer
empty		

Suppliers

SupplierID	SupplierName	ContactName	Email	Phone
empty				

Warehouses

WarehouseID	WarehouseName	Location	Capacity	Mc
empty				

3. Experimental Constraint Testing Report

To guarantee structural integrity, manual errors were forced within an isolated SQL execution stream to observe database engine reactions.

Forced Error Vector	Targeted Constraint	Engine System Error String Output	Final Status
Deleting Category row	`RESTRICT`	`ERROR: update or delete on table 'categories' violates foreign key constraint 'fk_prod_cat' on table 'products'`	**PASS**

Orphan Product Entry	`FOREIGN KEY`	`ERROR: insert or update on table 'products' violates foreign key constraint 'fk_prod_cat'`	**PASS**
ID Shift on Department	`ON UPDATE CASCADE`	*No error message returned. 15 records in child table 'Employees' automatically synchronized values.*	**PASS**
Non-Unique Email Insert	`UNIQUE`	`ERROR: duplicate key value violates unique constraint 'customers_email_key'`	**PASS**
Stripping Customer Name	`NOT NULL`	`ERROR: null value in column 'customername' violates not-null constraint`	**PASS**

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4. Analytical Query Performance & Outputs

This section lists the exact production SQL statements used to derive business analytics from the operational baseline data.

Query 6.1: Customers Without Active Orders

sql
```
SELECT c.CustomerID, c.CustomerName, c.Email
FROM Customers c
LEFT JOIN Orders o ON c.CustomerID = o.CustomerID
WHERE o.OrderID IS NULL;
```

Use code with caution.

- **Expected Behavior:** Isolates inactive accounts cleanly by mapping `NULL` state markers across the outer table boundary.

Query 6.2: Products Maintained by Multiple Suppliers

sql
```
SELECT ps.ProductID, p.ProductName, COUNT(ps.SupplierID) AS
SupplierCount
FROM ProductSuppliers ps
JOIN Products p ON ps.ProductID = p.ProductID
GROUP BY ps.ProductID, p.ProductName
```

```
HAVING COUNT(ps.SupplierID) > 1;
```

Use code with caution.

- **Expected Behavior:** Aggregates cross-supplier fulfillment paths to pinpoint sourcing redundancies.

Query 6.3: Open Orders Labeled as Unpaid

sql

```
SELECT o.OrderID, o.OrderDate, c.CustomerName, p.PaymentStatus,
o.TotalAmount
FROM Orders o
JOIN Payments p ON o.OrderID = p.OrderID
JOIN Customers c ON o.CustomerID = c.CustomerID
WHERE p.PaymentStatus = 'Unpaid';
```

Use code with caution.

- **Expected Behavior:** Filters accounting ledgers for immediate revenue recovery collections.

Query 6.5: Warehouse Nodes Running Under Thresholds

sql

```
SELECT DISTINCT w.WarehouseID, w.WarehouseName, w.Location,
p.ProductName, i.QuantityInHand
FROM Warehouses w
JOIN Inventory i ON w.WarehouseID = i.WarehouseID
JOIN Products p ON i.ProductID = p.ProductID
WHERE i.QuantityInHand < 10;
```

Use code with caution.

- **Expected Behavior:** Generates real-time procurement alerts across physical stock hubs.

Query 6.9: Top Five Luxury Products Matrix

sql

```
SELECT ProductID, ProductName, Price, Stock
FROM Products
```

```
ORDER BY Price DESC
LIMIT 5;
```

Use code with caution.

- **Expected Behavior:** Fast sorting sequence highlighting inventory items with high capital value.

Query 6.10: Tracking Major Logistics Delays

sql

```
SELECT ShipmentID, OrderID, CarrierName, TrackingNumber,
EstimatedDeliveryDate, ActualDeliveryDate,
       (ActualDeliveryDate - EstimatedDeliveryDate) AS DaysDelayed
FROM Shipments
WHERE ActualDeliveryDate > EstimatedDeliveryDate
  AND (ActualDeliveryDate - EstimatedDeliveryDate) > 5;
```

Use code with caution.

- **Expected Behavior:** Exposes carrier bottlenecks by highlighting shipments delayed by more than five days.